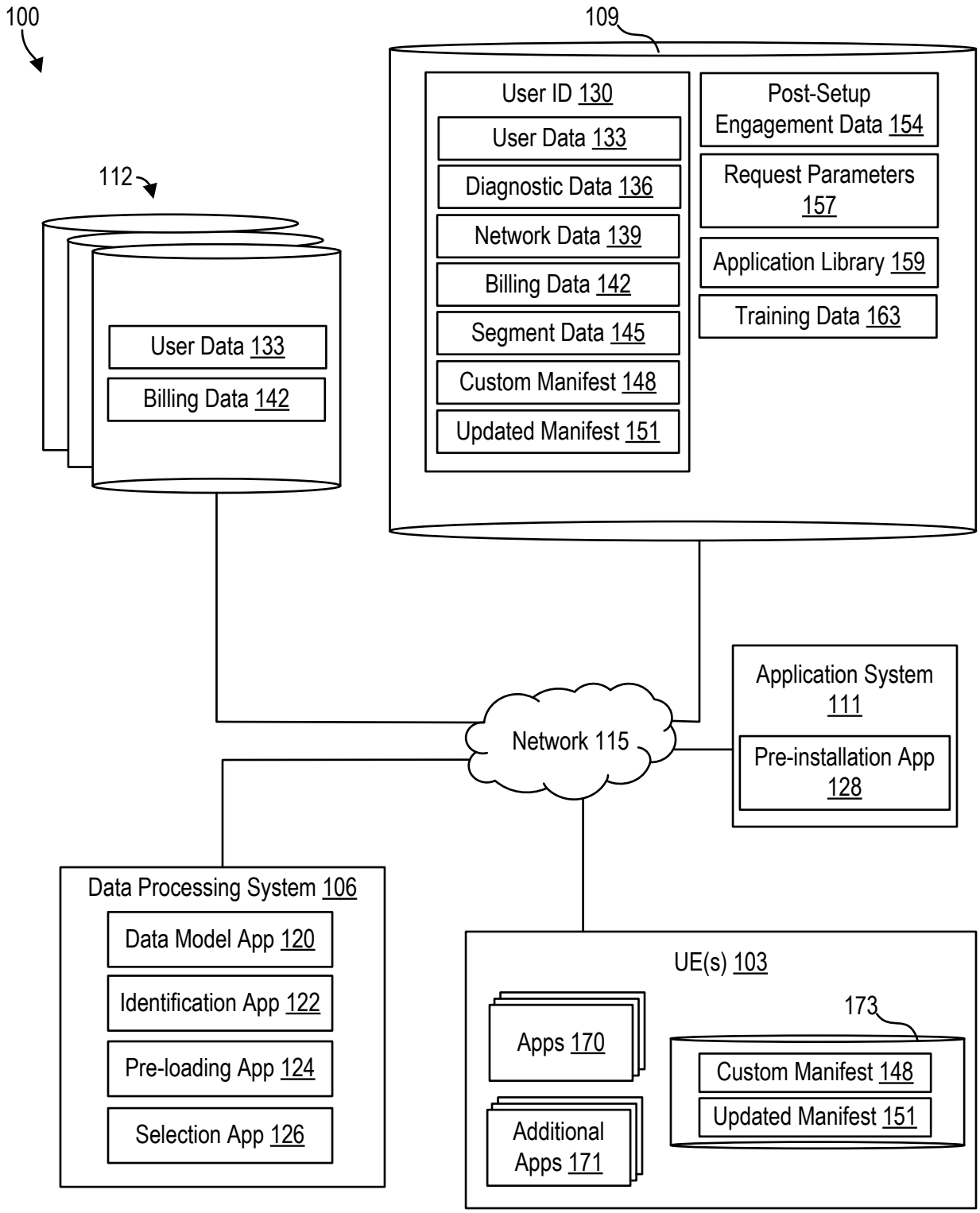
100
109
User ID 130
User Data 133
Diagnostic Data 136
Network Data 139
Billing Data 142
Segment Data 145
Custom Manifest 148
Updated Manifest 151
Post-Setup Engagement Data 154
Request Parameters 157
Application Library 159
Training Data 163
112
User Data 133
Billing Data 142
Network 115
Application System 111
Pre-installation App 128
Data Processing System 106
Data Model App 120
Identification App 122
Pre-loading App 124
Selection App 126
UE(s) 103
Apps 170
Additional Apps 171
173
Custom Manifest 148
Updated Manifest 151

FIG. 1

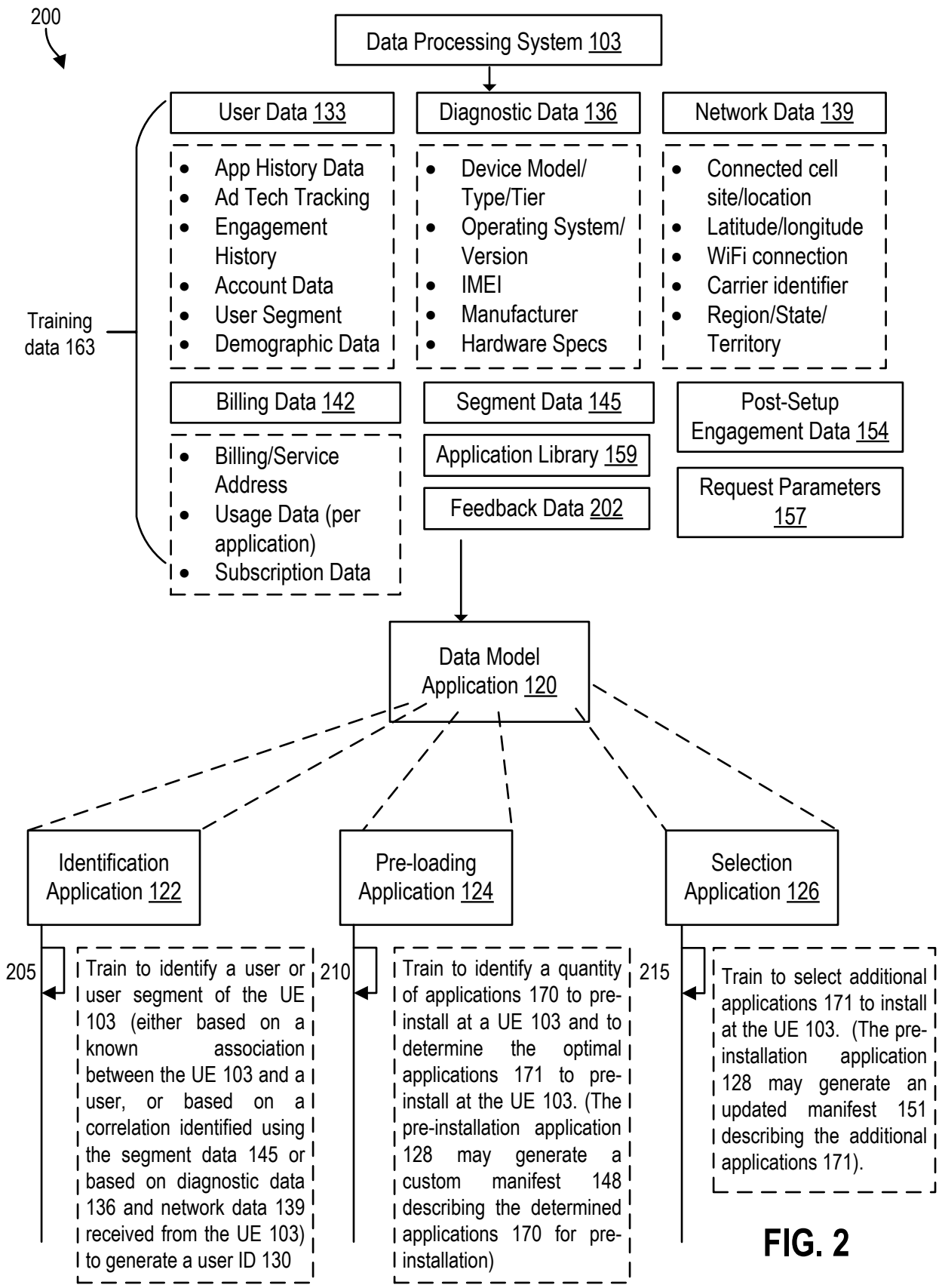
200
Data Processing System 103
User Data 133
Diagnostic Data 136
Network Data 139
App History Data
Ad Tech Tracking
Engagement History
Account Data
User Segment
Demographic Data
Device Model/ Type/Tier
Operating System/ Version
IMEI
Manufacturer
Hardware Specs
Connected cell site/location
Latitude/longitude
WiFi connection
Carrier identifier
Region/State/ Territory
Training data 163
Billing Data 142
Segment Data 145
Post-Setup Engagement Data 154
Billing/Service Address
Usage Data (per application)
Subscription Data
Application Library 159
Feedback Data 202
Request Parameters 157
Data Model Application 120
Identification Application 122
Pre-loading Application 124
Selection Application 126
205
Train to identify a user or user segment of the UE 103 (either based on a known association between the UE 103 and a user, or based on a correlation identified using the segment data 145 or based on diagnostic data 136 and network data 139 received from the UE 103) to generate a user ID 130
210
Train to identify a quantity of applications 170 to pre-install at a UE 103 and to determine the optimal applications 171 to pre-install at the UE 103. (The pre-installation application 128 may generate a custom manifest 148 describing the determined applications 170 for pre-installation)
215
Train to select additional applications 171 to install at the UE 103. (The pre-installation application 128 may generate an updated manifest 151 describing the additional applications 171).

FIG. 2

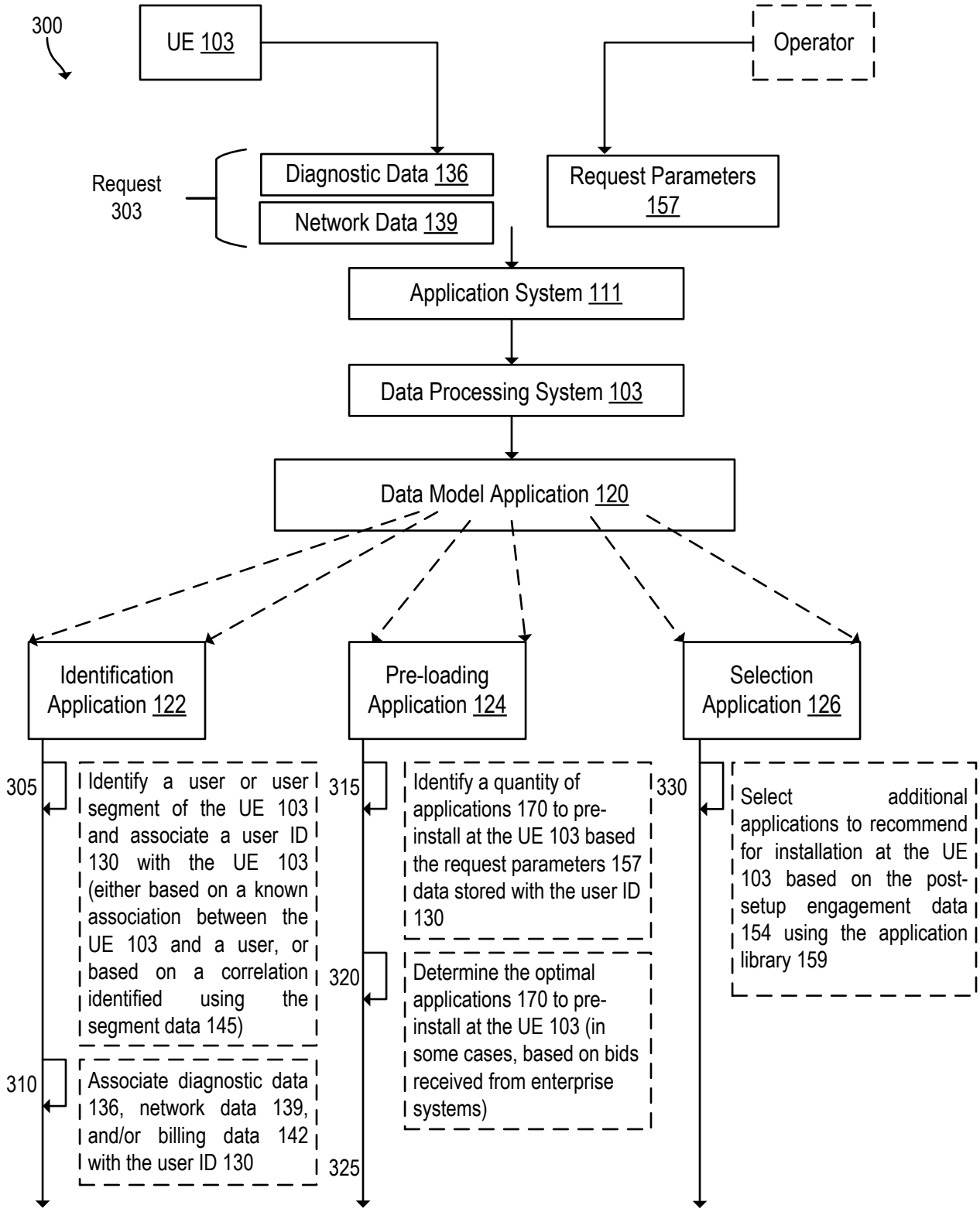
300
UE 103
Operator
Request 303
Diagnostic Data 136
Network Data 139
Request Parameters 157
Application System 111
Data Processing System 103
Data Model Application 120
Identification Application 122
Pre-loading Application 124
Selection Application 126
305
Identify a user or user segment of the UE 103 and associate a user ID 130 with the UE 103 (either based on a known association between the UE 103 and a user, or based on a correlation identified using the segment data 145)
310
Associate diagnostic data 136, network data 139, and/or billing data 142 with the user ID 130
315
Identify a quantity of applications 170 to pre-install at the UE 103 based the request parameters 157 data stored with the user ID 130
320
Determine the optimal applications 170 to pre-install at the UE 103 (in some cases, based on bids received from enterprise systems)
325
330
Select additional applications to recommend for installation at the UE 103 based on the post-setup engagement data 154 using the application library 159

FIG. 3

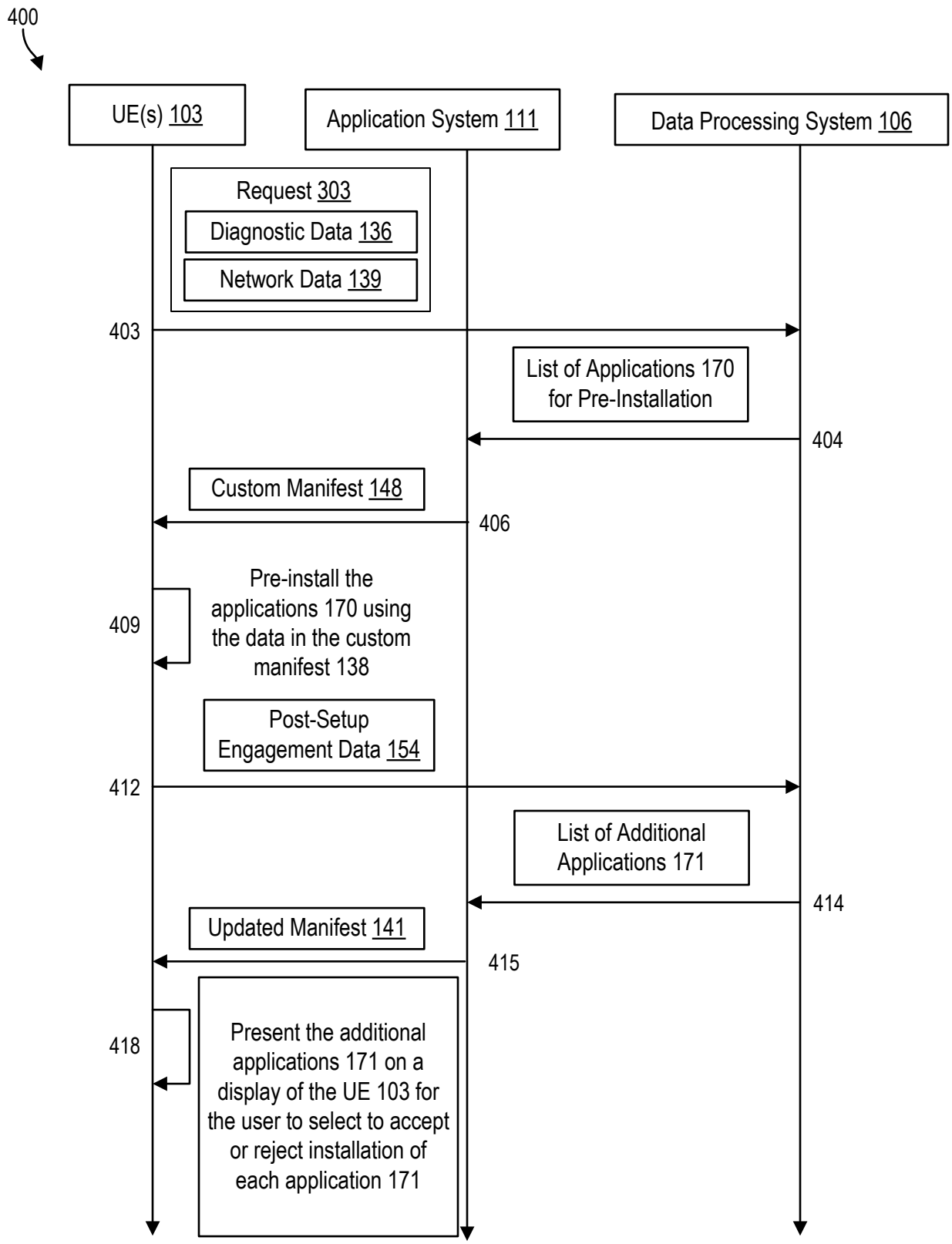
400
UE(s) 103
Application System 111
Data Processing System 106
Request 303
Diagnostic Data 136
Network Data 139
403
List of Applications 170 for Pre-Installation
404
Custom Manifest 148
406
Pre-install the applications 170 using the data in the custom manifest 138
409
Post-Setup Engagement Data 154
412
List of Additional Applications 171
414
Updated Manifest 141
415
Present the additional applications 171 on a display of the UE 103 for the user to select to accept or reject installation of each application 171
418

FIG. 4

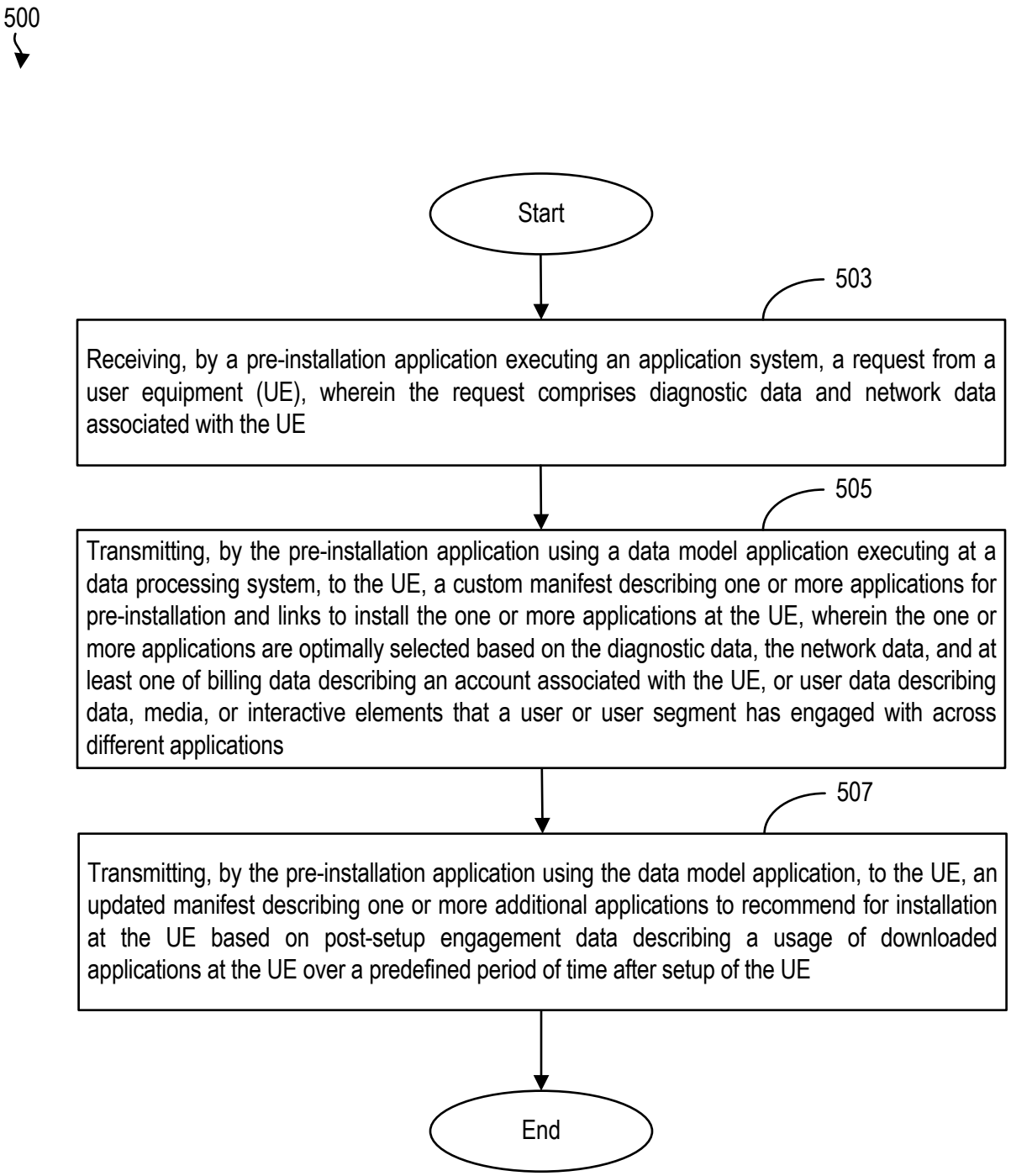
500
Start
503
Receiving, by a pre-installation application executing an application system, a request from a user equipment (UE), wherein the request comprises diagnostic data and network data associated with the UE
505
Transmitting, by the pre-installation application using a data model application executing at a data processing system, to the UE, a custom manifest describing one or more applications for pre-installation and links to install the one or more applications at the UE, wherein the one or more applications are optimally selected based on the diagnostic data, the network data, and at least one of billing data describing an account associated with the UE, or user data describing data, media, or interactive elements that a user or user segment has engaged with across different applications
507
Transmitting, by the pre-installation application using the data model application, to the UE, an updated manifest describing one or more additional applications to recommend for installation at the UE based on post-setup engagement data describing a usage of downloaded applications at the UE over a predefined period of time after setup of the UE
End

FIG. 5

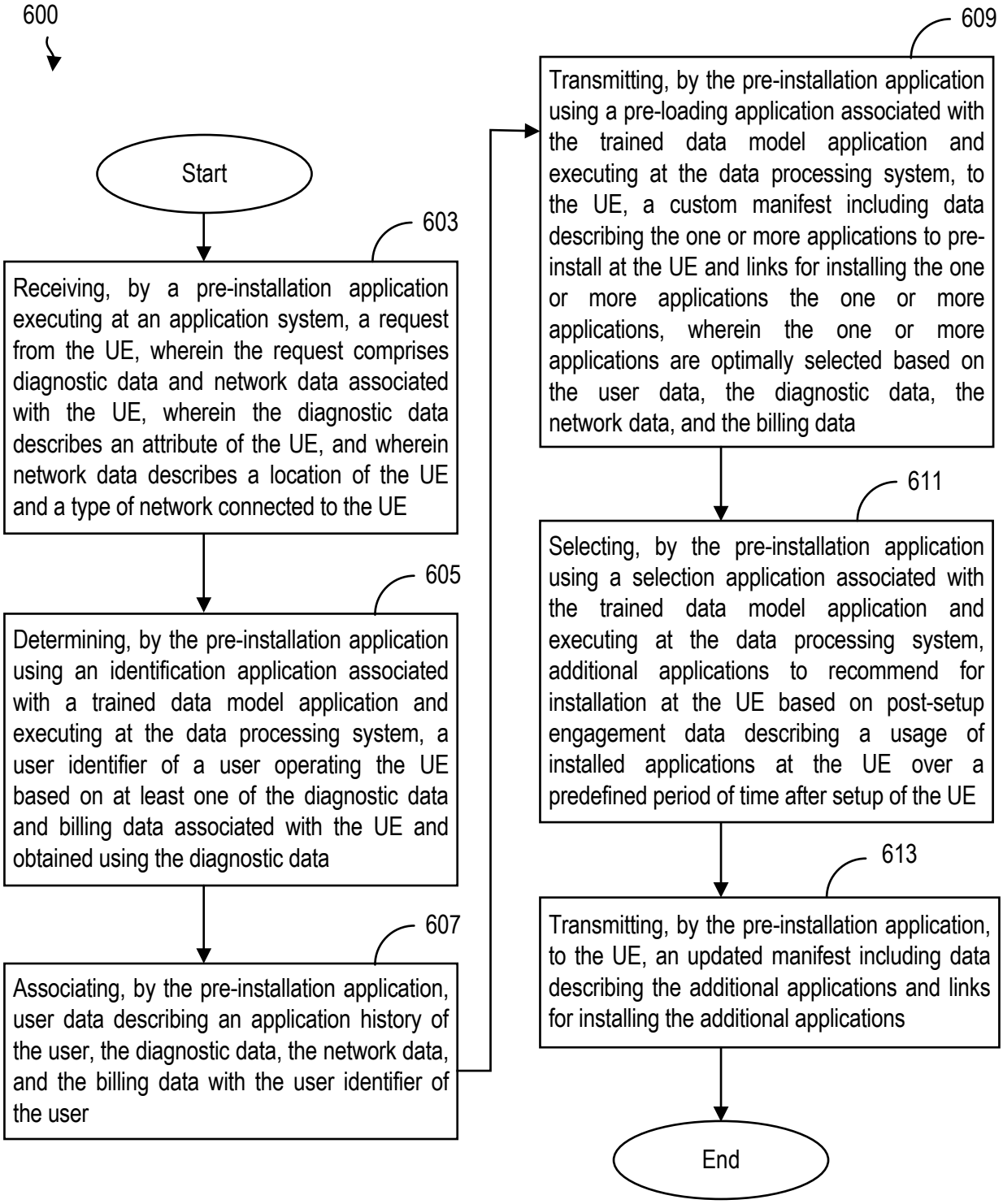
600
Start
603
Receiving, by a pre-installation application executing at an application system, a request from the UE, wherein the request comprises diagnostic data and network data associated with the UE, wherein the diagnostic data describes an attribute of the UE, and wherein network data describes a location of the UE and a type of network connected to the UE
605
Determining, by the pre-installation application using an identification application associated with a trained data model application and executing at the data processing system, a user identifier of a user operating the UE based on at least one of the diagnostic data and billing data associated with the UE and obtained using the diagnostic data
607
Associating, by the pre-installation application, user data describing an application history of the user, the diagnostic data, the network data, and the billing data with the user identifier of the user
609
Transmitting, by the pre-installation application using a pre-loading application associated with the trained data model application and executing at the data processing system, to the UE, a custom manifest including data describing the one or more applications to pre-install at the UE and links for installing the one or more applications the one or more applications, wherein the one or more applications are optimally selected based on the user data, the diagnostic data, the network data, and the billing data
611
Selecting, by the pre-installation application using a selection application associated with the trained data model application and executing at the data processing system, additional applications to recommend for installation at the UE based on post-setup engagement data describing a usage of installed applications at the UE over a predefined period of time after setup of the UE
613
Transmitting, by the pre-installation application, to the UE, an updated manifest including data describing the additional applications and links for installing the additional applications
End

FIG. 6

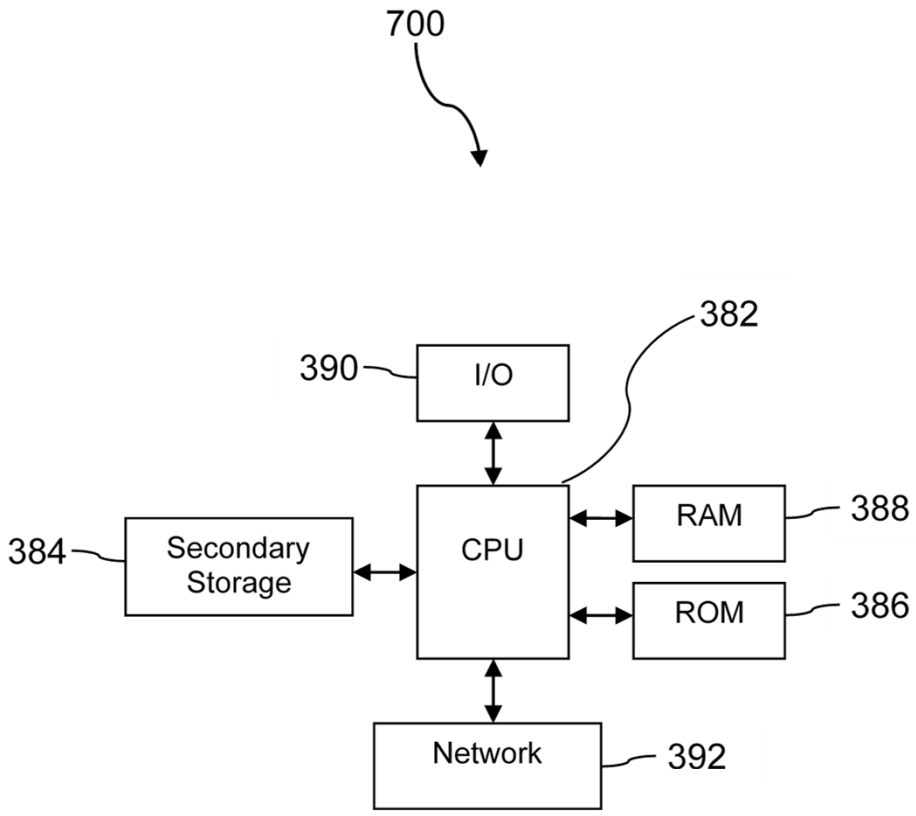
700
382
390
I/O
384
Secondary Storage
CPU
RAM
388
ROM
386
Network
392

FIG. 7